



HIGH-ALTITUDE WEATHER BALLOON PROJECT

Comprehensive combined technical and planning report for the 10 km validation, 20 km intermediate, and 35 km near-space mission tiers.



BEYOND THE BALLOON.

IN-SPARX is not designed as a one-time balloon demonstration. It is conceived as a student-led near-spaceresearch platform: a system that flies, measures, validates, recovers and learns.

WEATHER BALLOON

THE LAUNCH PROCESS

FROM GROUND TO HIGH ALTITUDE

SAME BALLOON.
SAME SCIENCE.
A HIGHER PERSPECTIVE.

01 PREPARE THE EQUIPMENT

The balloon, payload (radiosonde), parachute and all components are checked and assembled.



02 INFLATE THE BALLOON

The balloon is filled with helium (or hydrogen) until it reaches the right size and is ready for launch.



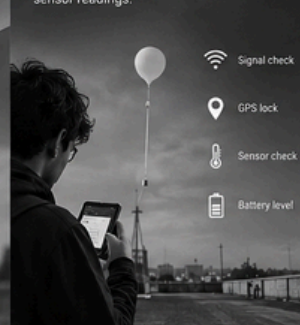
03 ATTACH THE PAYLOAD

The radiosonde (with sensors) is securely attached to the balloon, along with the parachute and support lines.



04 FINAL CHECKS

The system is tested for proper communication, GPS signal and sensor readings.



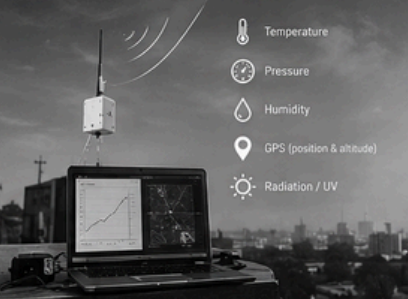
05 LAUNCH

The balloon is released and begins its ascent, carrying the payload into the atmosphere.



06 DATA COLLECTION

As the balloon rises, the radiosonde transmits real-time data (temperature, pressure, humidity, GPS, etc.) to a ground station.



07 BURST & DESCENT

At high altitude, the balloon expands and eventually bursts. The parachute then slows the payload's descent back to Earth.



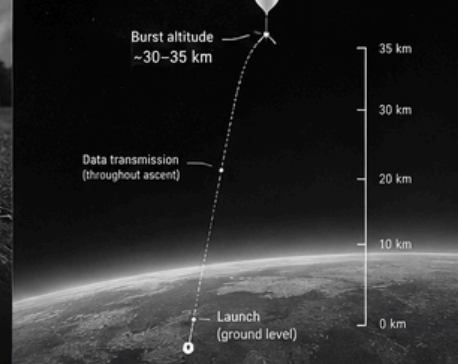
08 RECOVERY

The payload lands, is located (using GPS), and the data is downloaded for analysis.



THE JOURNEY

HIGHER. FARTHER. MORE DATA.



WEATHER BALLOON // ATMOSPHERE // DATA // RESEARCH // A CLEANER PLANET

OBSERVE • MEASURE • UNDERSTAND

THE CORE IDEA

SCIENCE + SYSTEM + REPEATABILITY

A flight is successful only when the observations are scientifically interpretable, the engineering chain is validated, and the data survives the mission.

A REUSABLE RESEARCH PLATFORM

The unique goal of IN-SPARX is to create a repeatable, student-built near-space research architecture—not merely to reach a high altitude once.

1

BUILD THE PLATFORM

Create an integrated payload architecture covering science instrumentation, avionics, telemetry, local logging and recovery.

2

PROVE IT STEP BY STEP

Use 10 km, 20 km and 35 km tiers as progressive validation points, increasing altitude only after the integrated system is demonstrated.

3

TURN FLIGHTS INTO EVIDENCE

Preserve calibrated measurements, raw data, metadata and flight records so every mission becomes a research asset and a design input for the next.

THIS CHANGES THE QUESTION

Instead of asking only “Can we send a payload to near-space?”, IN-SPARX asks “Can we build a reliable scientific measurement system that can repeatedly operate there?”

THE DIFFERENCE

INTEGRATION IS THE INNOVATION

The individual technologies are established. The proposed research value lies in integrating them into a staged, traceable workflow where science and engineering validate each other.

1

SCIENCE PAYLOAD

Temperature, pressure, humidity, UV, radiation, motion and GPS can form a vertically resolved observational package.

2

FLIGHT SYSTEM

Power, thermal protection, flight computer, telemetry and recovery are validated as one system.

3

DATA INTEGRITY

Local logging protects the primary dataset when the communications link is interrupted.

4

MEASUREMENT TRACEABILITY

Calibration, sensor limits, timestamps, configuration and uncertainty remain attached to the observations.

THE UNIQUE COMBINATION

A balloon becomes a research platform only when the measurement chain is understood from sensor to recovered dataset. IN-SPARX treats that entire chain as the object of design.

**QUESTION**

Identify the research goal, define the problem and formulate key questions.

**INSTRUMENT**

Design and integrate the sensors, payload and systems for the mission.

**FLIGHT**

Launch the balloon and execute the mission plan.

**DATA**

Collect, store and validate the data during and after the flight.

**EVIDENCE**

Analyse the results, verify findings and generate research outcomes.

THE NEED

A MEASUREMENT IS NOT YET A DATASET

Near-space conditions create a demanding environment: cold, low pressure, changing RF conditions, motion and recovery uncertainty can all affect the validity of an observation.

1 ATMOSPHERIC CONTEXT

Temperature and pressure form the core atmospheric state variables; humidity, UV, radiation, motion and position add context.

2 INSTRUMENT LIMITS

Sensor range, calibration, installation effects and environmental exposure must be known before measurements can support scientific interpretation.

3 DATA SURVIVAL

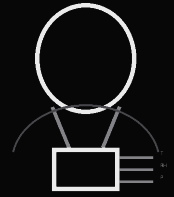
Local logging, timestamps and configuration metadata ensure that a communications interruption does not erase the primary scientific record.

THE RESEARCH NEED IN ONE LINE

Create measurements that are not only collected, but calibrated, traceable, comparable and repeatable.

FROM FLIGHT

TO RESEARCH



IN-SPARX is designed to produce both scientific observations and engineering evidence that can guide future missions.

1**ATMOSPHERIC OBSERVATION**

Altitude-resolved in-situ profiles can be compared with standard-atmosphere references and, where available, independent upper-air observations.

2**MEASUREMENT SCIENCE**

Calibration, operating limits, installation effects and uncertainty are treated as part of the scientific result.

3**NEAR-SPACE ENGINEERING**

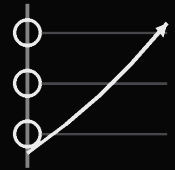
Cold, power, RF, telemetry, logging and recovery tests create evidence about the performance of an integrated flight system.

4**REUSABLE KNOWLEDGE**

Raw data, configuration records, flight logs and anomalies remain available for analysis and the next mission.

The intended contribution is therefore cumulative: each mission adds observations, identifies limitations, and reduces uncertainty for the next iteration.

ONE SYSTEM. THREE TIERS. ONE RESEARCH LOOP.



10 km → validate the chain | 20km → extend exposure and redundancy | 35 km → run the full science configuration

10 km

Power · sensors ·
GPS · telemetry ·
logging ·
recovery

20 km

Cold exposure ·
redundancy ·
tracking · data
integrity

35 km

Full science
payload · RF
margin · thermal
margin · traceable
data

THE LOGIC

Altitude is not the achievement by itself. The achievement is a validated system that can produce scientifically defensible evidence as altitude and environmental complexity increase.

IN-SPARX

INSTRUMENTATION FOR NEAR SPACE PHYSICS,
AEROSPACE RESEARCH AND EXPLORATION

THE PROPOSAL

IN-SPARX proposes a staged, student-led near-space research program in which high-altitude balloon flights serve simultaneously as scientific observation platforms and engineering validation experiments.

Its unique goal is to develop a reusable mission architecture capable of producing reliable, traceable atmospheric measurements while progressively validating the avionics, telemetry, thermal protection, data logging and recovery chain.

SUCCESS IS MEASURED BY EVIDENCE

Calibrated data · validated instrumentation · reliable telemetry · preserved raw records · successful recovery · documented uncertainty

THE QUESTION IN FRONT OF THE TEAM

Can we build a near-space platform that does not end with a flight, but becomes a repeatable scientific research capability?



About this report

This report merges every IN-SPARX source document produced so far into a single, non-redundant technical and planning reference, organized around seven core sections requested for the project: Aim & Objective, Required Equipment, Component Specifications, Cost Estimation, Phases of the Project, Testing Procedure, and Workflow. Detailed scientific background, flight-physics calculations, team structure and full source listings are preserved in the Appendices so that no technical detail from the source material is lost.

Cost figures throughout this report are planning estimates synthesized from student-club budgets and component price lists supplied in the source material — they are not vendor quotations and should be confirmed with current Indian suppliers before purchase. Team-size recommendations are practical student-project inferences, not a universal industry average.

SOURCE / PROJECT BASIS

MIT-WPU Student-Led Near-Space Research Initiative • September 2026

Contents

01	Aim and objective
02	Required equipment
03	Component specifications
04	Cost estimation
05	Phases of the project
06	Testing procedure
07	Workflow
A-E	Appendices A-E: atmosphere, flight physics, team structure, technology roadmap, and source documents



04 - Aim and objective

A climate-facing mission needs a measurable question, a traceable measurement chain, and a defined validation plan.

MISSION AIM

Design, ground-test, launch and recover a staged series of high-altitude balloon payloads at nominal 10 km, 20 km and 35 km tiers. The scientific objective is to obtain vertically resolved in-situ measurements of atmospheric state and selected radiation/optical variables while simultaneously validating a modular avionics, telemetry, logging and recovery system.

WHAT IS MEASURED

Temperature and pressure form the core atmospheric state variables. Humidity, UV, radiation, motion and position add context. GPS provides altitude independent of the pressure-sensor ceiling. Time-stamped local logging preserves the data when the radio link is unavailable.

CORE RESEARCH QUESTIONS

EXTENDED RESEARCH FRAMEWORK

- 01 How do temperature, pressure and humidity evolve with altitude during ascent and descent, and where do the profiles depart from expected atmospheric structure?
- 02 What systematic temperature biases can arise from solar loading, sensor radiation balance, ventilation, balloon wake and sensor placement?
- 03 How large are humidity response errors caused by time lag, cold/dry conditions, contamination or reduced sensor sensitivity?
- 04 Can independent measurements distinguish true atmospheric variability from instrument-induced variability?
- 05 How does sensor performance change across the 10 km, 20 km and 35 km validation tiers as temperature and pressure decrease?
- 06 How much does the physical installation?shielding, airflow, self-heating, vibration and enclosure design?change the measured signal?

- 07 Can GPS altitude and pressure-derived altitude be cross-checked to identify the pressure sensor?s valid operating region and detect drift or saturation?
- 08 Can paired ascent/descent observations reveal response-time effects, hysteresis or other non-equilibrium behaviour in temperature and humidity sensors?
- 09 What calibration and correction functions are justified by co-measurement data, and how stable are those corrections across flights and conditions?
- 10 How should uncertainty be propagated from calibration, sensor response, installation effects and environmental exposure into the final profile?
- 11 Which differences between IN-SPARX and radiosonde/upper-air data are explainable by known sensor characteristics, and which require further investigation?
- 12 Can a repeatable measurement-and-correction workflow turn each flight into a comparable research dataset rather than an isolated sounding?

RESEARCH LOGIC

Measure ? calibrate ? characterize bias ? cross-check ? correct only when justified ? quantify uncertainty ? compare across flights.

MISSION INTERPRETATION

The 10 / 20 / 35 km values remain staged validation targets, not assumed atmospheric states. Actual profiles depend on season, weather, location and flight conditions.

Radiosondes are powerful · not bias-free

Two established studies show why upper-air observations must be interpreted with sensor physics, calibration history and environmental effects in view.

THE STARTING POINT

Radiosondes are a traditional, high-value method for vertical atmospheric profiling. But the sensor reading is not automatically identical to the atmospheric state: radiation, ventilation, response time, contamination, temperature dependence and instrument changes can introduce systematic bias.

01 TEMPERATURE MEASUREMENT

Use of Radiosonde Temperature Data in Climate Studies

James K. Luers & Robert E. Eskridge · Journal of Climate 11 (1998), 1002-1019

Luers and Eskridge evaluated ten widely used radiosonde systems and showed that sensor temperature can differ from air temperature under different environmental conditions. Important sensitivities included solar angle, cloud cover, ambient temperature, blackbody temperature and ventilation velocity. They also show that correction models can improve climate use of some sonde records.

IN-SPARX implication Treat temperature as a measurement problem, not just a sensor output. IN-SPARX therefore tests placement, shielding, airflow/ventilation, thermal behaviour and calibration before interpreting the profile.

02 HUMIDITY MEASUREMENT

Performance of operational radiosonde humidity sensors in direct comparison with a chilled mirror dew-point hygrometer and its climate implication

Junhong Wang et al. · Geophysical Research Letters 30 (2003), 1860

Wang et al. compared operational radiosonde humidity sensors with a research-quality chilled-mirror hygrometer. The study found that the Vaisala RS80-H agreed well in the lower and middle troposphere but developed a dry bias in the upper troposphere, while Sippican carbon-hygristor sensors showed time-lag errors and loss of response to humidity changes at cold temperatures. The authors show that sensor changes can create artificial shifts in long-term humidity records.

THE IN-SPARX RESPONSE: MEASURE THE BIAS, DON'T HIDE IT

01 CALIBRATE	Traceable offsets, response, calibration date and uncertainty are recorded before integration.	04 REFERENCE-COMPARISON	Where access permits, co-launch/co-locate with an independent upper-air reference and quantify differences.
02 CHARACTERIZE	Test radiation, airflow, self-heating, cold exposure, response time and sensor range.	05 CORRECT + REPORT	Use a correction only when evidence supports it; preserve raw, corrected and uncertainty information.
03 CROSS-CHECK	Compare GPS vs pressure altitude, independent temperature observations and local logs vs telemetry.	KEY PRINCIPLE	IN-SPARX makes the measurement chain explicit and tests when a correction is justified.

REFERENCES Luers & Eskridge (1998), Journal of Climate, 11, 1002-1019. Wang et al. (2003), Geophysical Research Letters, 30, 1860. DOI: 10.1175/1520-0442(1998)011<1002:UORTDI>2.0.CO;2 · 10.1029/2003GL016985

05 -

Three tiers. One system.

Each tier is a validation step: altitude increases only after the integrated payload, tracking and recovery chain has been demonstrated.

Mission parameter	Tier 1 / 10 km	Tier 2 / 20 km	Tier 3 / 35 km
Atmospheric regime	Troposphere / near-tropopause	Lower-mid stratosphere	Stratosphere / near-space research tier
ISA reference*	-50.0 °C / 264 hPa	-56.5 °C / 54.7 hPa	-36.1 °C / 5.6 hPa
Balloon class	Illustrative 350-500 g latex	Illustrative 800-1200 g latex	Illustrative 1200 g+ latex class
Example payload	MCU + GPS + LoRa + logging	MCU + LoRa/APRS + environmental sensors	Full avionics + redundant tracking + science payload
Thermal approach	Insulated payload enclosure	Insulation + tested supplemental heat	Insulation + thermal-control strategy verified by test
Recovery / tracking	GPS + LoRa + audible locator	LoRa + APRS + visual locator	LoRa + APRS + independent backup tracking

1

10 km

Validation flight

Demonstrate the complete payload chain at a lower target altitude: power, sensors, GPS, RF telemetry, local logging and recovery. The purpose is system validation, not a claim that the atmosphere is identical at every 10 km launch.

02

20 km

Intermediate flight

Increase stratospheric exposure and add tracking/thermal redundancy. Use the flight to quantify cold-soak behaviour, telemetry continuity and sensor performance before the 35 km campaign.

03

35 km

Near-space research tier

Operate the full science configuration. Treat pressure-sensor range, thermal margins, RF link margin, recovery and calibration as explicit acceptance criteria rather than assumptions.

Required equipment

Equipment is grouped by subsystem. Every item below appears in at least one of the underlying IN-SPARX documents; items usable at more than one altitude tier are marked accordingly in Section 3.

2.1 FLIGHT VEHICLE & LIFT

- Latex weather balloon — 350 g to 2000 g class depending on tier (zero-pressure, open-duct type for standard sounding flights).
- Lifting gas: helium (hydrogen only where explicitly permitted and handled safely).
- Recovery parachute — 4–6 ft ripstop nylon canopy.
- Automatic cutdown / nichrome hot-wire line burner.
- High-tensile braided nylon rigging cordage / flight-train harness (50 kg rated).
- Optional: superpressure envelope for long-duration constant-altitude float missions.

2.2 STRUCTURE & THERMAL PROTECTION

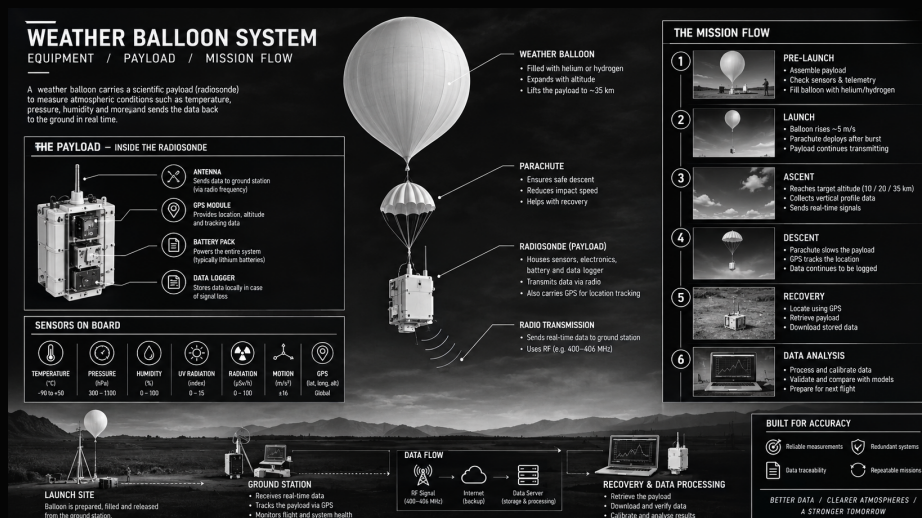
- Payload enclosure — 25 mm XPS or high-density EPS foam cube.
- External Mylar reflective coating for passive solar-radiation shielding.
- Internal chemical hand warmers / active heating element for the 35 km tier.
- Ventilated, 3D-printed Stevenson-screen-style radiation shield (PETG/ABS) for the external temperature probe.

2.3 AVIONICS: FLIGHT COMPUTER & POWER

- Flight computer: dual-core ESP32, STM32, Arduino Nano 33 BLE, or Teensy 4.1.
- Power bank: dual Li-FeS2 Energizer Ultimate Lithium primary cells and/or 3.7 V LiPo pack with 5 V boost converter.
- High-efficiency LDO voltage regulators with stable 3.3 V / 5.0 V rails.
- Custom PCB with JST-XH locking connectors to eliminate vibration-induced disconnects.
- MicroSD card data logger (SPI, Class 10, 16 GB+, FAT32).
- Hardware watchdog timer (WDT) for automatic soft-lock recovery.

2.4 TELEMETRY & RF

- Primary: SX1276 LoRa, 434 MHz or 868 MHz ISM, Chirp Spread Spectrum.
- Secondary: 144.390 MHz APRS VHF transmitter using AX.25 packets.
- Satellite backup: SPOT Trace or Iridium SBD module.
- Independent 433 MHz CW beacon; quarter-wave payload monopole; 5-element Yagi + LNA; ESP32 ground receiver.



02

The payload stack

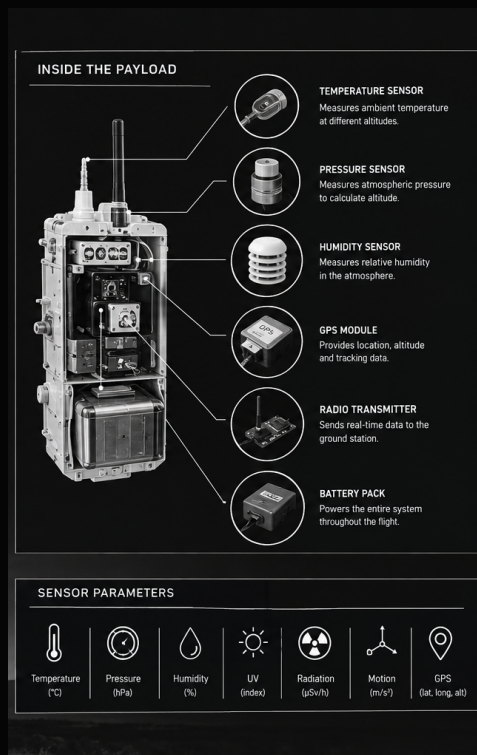
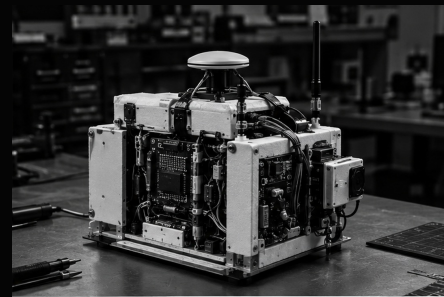
Science payload, recovery hardware, ground station and field operations complete the system around the core avionics.

2.5 INSTRUMENTATION / SCIENCE PAYLOAD

- Pressure / temperature / humidity: Bosch BME280 or BMP388.
- External air temperature: waterproof DS18B20 probe.
- Motion / spin-rate: ICM-20948 9-DOF IMU.
- Cosmic-ray radiation: CJMCU-1171 or J305 Geiger-Müller tube + boost converter.
- UV irradiance: VEML6075 (UV-A / UV-B).
- Air quality: SDS011 particulate sensor and/or NDIR CO2 sensor.
- Positioning: u-blox MAX-M8Q or NEO-M8N GPS, Airborne <1g / Flight Mode 6.
- Optional: RunCam or modified GoPro for visual ground-truth imagery.

2.6 RECOVERY & LOCATOR HARDWARE

- 100 dB piezo buzzer, triggering below ~1000 m descent altitude.
- High-intensity flash LED / strobe for low-light recovery.
- Independent CW beacon for final-approach direction finding.



2.7 GROUND STATION & SOFTWARE

- Field laptop / ground computer; Node-RED or Python LoRa/APRS packet decoder.
- Grafana live dashboard for altitude, ascent rate and battery voltage.
- Leaflet JS map overlay for predicted vs. actual GPS flight path.
- CUSF Tawhiri flight-path predictor for wind-drift and landing-zone simulation.
- FreeRTOS embedded firmware framework; RadioLib for SX1276 configuration.

2.8 FIELD & OPERATIONS EQUIPMENT

- Digital scale for precision helium-fill / free-lift measurement.
- Mobile recovery / chase vehicle with handheld Yagi direction finder.
- Printed launch checklist, NOTAM paperwork and handheld wind meter.

08 —

Component specifications

The core avionics architecture combines flight computer, sensor buses, watchdog recovery, local logging, long-range telemetry and independent tracking redundancy.

Component	Specification	Role	Component	Specification	Notes
ESP32 (dual-core)	240 MHz, Wi-Fi/BLE SoC	Sensor polling via I2C/SPI, low-power modes, non-blocking flight-state machine	SX1276 LoRa module	434 MHz or 868 MHz ISM, CSS modulation, up to +20 dBm Tx	Datasheet-class receive sensitivity ≈ -132 dBm at 125 kHz bandwidth / Spreading Factor 10; >100 km line-of-sight range
STM32	ARM Cortex-M series MCU	Alternate flight computer for the 35 km-class payload	APRS VHF	144.390 MHz, AX.25 packets	Received by civilian digipeaters and the APRS.fi network
Arduino Nano 33 BLE	ARM Cortex-M4 + integrated LSM9DS1 9-axis IMU	Compact flight computer used in the LoRa Telemetry Master Plan	SatCom backup	SPOT Trace / Iridium SBD	Global coverage independent of line-of-sight, for the 35 km tier
Teensy 4.1	High-performance ARM MCU	Alternate flight computer option (Flight Vehicle & Recovery spec)	Payload antenna	Quarter-wave wire monopole	VSWR < 1.5
Hardware WDT	Watchdog timer	Automatic recovery from software soft-locks	Ground antenna	5-element Yagi + LNA	$\sim +9$ dBi gain (used directly in the link-budget calculation, Appendix B)
Bus wiring	Isolated I2C & SPI with decoupling capacitors	Noise-tolerant multi-sensor bus	Beacon	433 MHz CW	Independent close-range direction finding

Component	Rating	Notes
Energizer Ultimate Lithium (Li-FeS2) AA	Operates reliably to -40°C	Chosen over alkaline/standard LiPo, which typically fail near -20°C
3.7 V LiPo + 5 V boost booster	~ 1800 mAh class	Cost-effective, but cold-sensitive: internal resistance rises sharply in the cold, and terminal voltage can sag under transmit-current load enough to trip the boost converter's under-voltage cutoff even with charge remaining
High-efficiency LDO regulators	Stable 3.3 V / 5.0 V rails	Powers MCU, radio and sensor bus

Cold-sag mitigation: the EPS/XPS thermal hull traps the small amount of waste heat the electronics themselves generate, keeping the internal bay meaningfully warmer than outside air through the tropopause crossing. See Appendix B, Section B.4, for the full analysis and recommended cold-chamber verification test.

09 -

Instrumentation has a ceiling

A climate dataset is only useful when the measurement range, calibration, exposure and uncertainty are explicit.

PRESSURE MEASUREMENT LIMIT

The Bosch BME280 has an operating pressure range of 300-1100 hPa. Under an ISA atmosphere, 300 hPa occurs at roughly 9.2 km. Above that pressure range, the sensor should not be treated as the primary source for quantitative pressure-altitude retrieval.

Instrument	Measured variable	Scientist-facing interpretation
BME280 / BMP388	Pressure, temperature, humidity	Use pressure only inside the device's specified pressure range; record raw output and sensor temperature.
DS18B20	External air temperature	Mount so the sensing element samples ambient air while minimizing radiative/self-heating bias.
ICM-20948	Acceleration / angular motion	Use for spin-rate and flight-dynamics context; validate orientation and sampling before flight.
UV sensor	UV-A / UV-B irradiance	Interpret as irradiance only after calibration and accounting for enclosure/window transmission.
Radiation detector	Count rate / particle events	Report counts with integration time, detector response and background characterization; counts are not automatically an absolute dose measurement.
GPS	Position / altitude / time	Primary altitude reference above the BME280 pressure ceiling; record fix quality and update rate.

HOW THE DATA SHOULD BE REPORTED

Every measurement should retain UTC time, GPS position/altitude, sensor identity, configuration, sampling rate and calibration information. Flag saturation, missing packets, sensor warm-up, ascent/descent phase and any period outside the specified operating range. This turns a sensor stream into a defensible atmospheric dataset.

10 -

Balloon sizing is the altitude decision

Burst altitude is an outcome of balloon geometry, fill state, free lift, payload mass, material properties and atmospheric conditions-not balloon mass alone.

PHYSICS IN ONE LINE

Net lift at release depends on displaced air mass minus the mass of helium, envelope, flight train and payload. The chosen free lift controls ascent rate; ascent rate and balloon expansion determine the time and altitude reached before burst.

Illustrative class	Planning burst diameter	Planning altitude band	Interpretation
200 g	~2.5-3.0 m	~18-22 km	Low-altitude validation only; supplier-specific.
350-450 g	~4.5-5.5 m	~25-28 km	Intermediate class; actual burst depends strongly on fill and payload.
600 g	~6.0-7.0 m	~28-32 km	Candidate for higher-altitude testing; verify manufacturer data.
800-1000 g	~7.5-8.5 m	~32-35 km	Candidate 35 km class; verify burst diameter and payload mass.
1200-1500 g	~9-10 m	~35-38 km	High-altitude class; requires supplier-specific sizing and trajectory analysis.

WHAT MUST BE MEASURED BEFORE FREEZING THE DESIGN

- Complete flight-train mass, including payload, parachute, rigging and trackers.
- Balloon model, nominal mass and manufacturer burst characteristics.
- Helium fill state and measured free lift using a calibrated scale.
- Predicted ascent rate and trajectory under forecast upper-air winds.
- Margin for temperature-dependent latex behaviour and manufacturing variability.

Scientist-facing note: the burst-altitude numbers above are planning bands, not guarantees. Use the supplier’s current datasheet and the actual filled-balloon mass/geometry for mission prediction.

11 — The mission becomes fundable

Cost bands, contingencies, and tier-wise planning envelopes turn the IN-SPARX concept into a repeatable student mission.

MISSION PLANNING

COST BANDS

CONTINGENCIES

TIER-WISE

PLANNING ENVELOPES



NEAR-SPACE BALLOON MISSION

12 — Four budgets, one planning model

STUDENT-CLUB CAP

\$265

Approximately ₹22,000 for a repeatable full mission.

TELEMETRY SUBSYSTEM

₹4,540

SX1276, Arduino Nano 33 BLE, BME280, MicroSD, and LiPo boost.

PLANNING SYNTHESIS

₹15,000–35,000

A recoverable multi-sensor flight with testing and contingency.

13 —

Stage the spend with the risk

PLANNING BANDS BY PROJECT STAGE

Stage	Planning envelope	Purpose
Bench prototype	₹3,000-8,000	Sensors, MCU, wiring, logging, ground tests
Recoverable student flight	₹15,000-35,000	Balloon, gas, payload, tracking, parachute, testing, operations
Robust / advanced mission	₹25,000-50,000+	Redundant tracking, thermal protection, radiation/UV, camera, spares

Planning bands by project stage

14 — The roadmap moves through review gates

01 Requirements and research

Define the question, sub-teams, and hardware architecture.

02 Bench prototype

Wire, flash, poll, log, and verify every sensor independently.

03 Environmental testing

Cold-chamber power test and 5 km+ RF line-of-sight test.

04 Design reviews

PDR, full mission rehearsal, CDR, then FRR and go/no-go.

15 -

Readiness before launch day

PRE-FLIGHT PRINCIPLE

Configure the payload as it will fly, run the complete electrical and communications chain, verify recovery hardware, confirm environmental limits, and rehearse the ground-station workflow before a flight decision is made.

1 PDR - PRELIMINARY DESIGN REVIEW

Preliminary Design Review. This is the first formal design checkpoint: it asks whether the mission has been defined clearly, the preliminary architecture is workable, and the important requirements have a credible verification path.

Purpose: establish that the preliminary design satisfies the mission requirements with acceptable risk and that the verification approach is credible.

Mission objectives; requirements; system architecture; interfaces; preliminary mass/power budget; RF (radio-frequency) link budget; thermal concept; hazards/risk register; preliminary test and calibration plan.

GATE OUTCOME Proceed to detailed design only when major requirements, interfaces and verification methods are defined and unresolved risks have owners.

2 CDR - CRITICAL DESIGN REVIEW

Critical Design Review. This is the detailed-design checkpoint: it asks whether the design is mature enough to fabricate, assemble, integrate and test as one controlled system.

Purpose: demonstrate that the design is mature enough to fabricate, assemble, integrate and test as a complete system.

Final schematics and mechanical layout; interface definitions; component/BOM (bill of materials) freeze; software configuration; power and thermal analysis; recovery design; telemetry plan; acceptance criteria; detailed verification procedures.

GATE OUTCOME Proceed to full integration/testing when the design is sufficiently closed and remaining changes are controlled and documented.

3 FRR - FLIGHT READINESS REVIEW

Flight Readiness Review. This is the final pre-launch readiness checkpoint: it asks whether the actual flight-configured system, ground system, and procedures are ready for launch under the current conditions.

Purpose: determine whether the configured flight system, ground system, people and procedures are ready for a safe launch and mission.

Flight-configured payload; completed functional tests; GPS (Global Positioning System) and RF checks; data logging; recovery-system checks; weather and upper-air assessment; airspace coordination/NOTAM (aeronautical notice) status; landing prediction; crew roles; checklists; open-risk disposition.

GATE OUTCOME GO only when required evidence is complete and launch constraints are satisfied; otherwise HOLD/NO-GO until the issue is resolved or formally accepted by the responsible authority.

DESIGN -> REHEARSAL -> RELEASE

PDR defines the problem and preliminary solution - CDR closes the design - FRR verifies flight readiness.

16 -

The first flight validates the chain

Workflow is staged so that altitude is added only after the payload, data and recovery chain has demonstrated acceptable performance.

1

10 km - VALIDATION FLIGHT

Goal: prove the integrated system.

Payload: core flight computer, GPS, environmental sensing, local logging and primary telemetry.

Acceptance focus: continuous power; valid time/position records; stable sensor sampling; telemetry reception; recovery-system function; no critical integration fault.

2

20 km - INTERMEDIATE FLIGHT

Goal: extend environmental exposure and redundancy.

Payload: retain the validated core while adding the planned secondary tracking and cold-protection measures.

Acceptance focus: thermal performance through the colder stratospheric segment; continuity of primary/secondary tracking; data integrity during link interruptions; recovery readiness.

3

35 km - NEAR-SPACE RESEARCH TIER

run the full science configuration at the target altitude.

Payload: full telemetry/redundancy architecture plus the selected radiation/UV and atmospheric measurements.

Acceptance focus: pressure-sensor ceiling handled correctly; GPS altitude retained; RF margin demonstrated; power/thermal margins remain positive; science data are calibrated and traceable.

The workflow is a verification ladder: lower-tier flights reduce uncertainty in integration, then higher tiers add altitude, cold exposure, range and scientific complexity.

17 -

Test the sensors against reality

Calibration is necessary but not sufficient: the payload must also be validated as an integrated measurement system in the environment it will experience.

1 CALIBRATE

Establish traceable offsets and response using an appropriate reference or manufacturer procedure before integration. Record calibration date, configuration and uncertainty.

2 VALIDATE RANGE

Confirm that each sensor remains within its specified operating range during the expected ascent/descent profile. Mark periods of saturation or out-of-range operation.

3 CHECK INSTALLATION

Verify airflow, radiation shielding, self-heating, mechanical vibration, electrical noise and enclosure effects. A good sensor can still produce biased data if its installation is poor.

4 CROSS-CHECK

Compare independent variables where possible: GPS altitude against pressure-derived altitude within the pressure-sensor range; external temperature against a second reference; packet telemetry against local SD logs.

5 RESERVE METADATA

Log timestamps, GPS position, sensor IDs, sample rates, firmware/configuration version, calibration coefficients and any known anomalies so the dataset can be reconstructed after flight.

MEASUREMENT QUALITY IS A SYSTEM PROPERTY

For a climate-science audience, report the measurement uncertainty and limitations alongside the plotted profile. Do not silently convert a sensor output into a scientific quantity that the device was not calibrated to provide.

18 -

Cold and RF are the two hard tests

The laboratory and ground tests should reproduce the coupled failure modes that are most likely to corrupt the flight dataset.

1

COLD / POWER

rated cold test

Operate the flight computer, power chain, sensors, logging and radio together while the enclosure is exposed to a controlled cold environment. Monitor battery voltage, current, resets, sensor output and data logging. The objective is to demonstrate margin, not merely that individual components work at room temperature.

2

RF / LINK

nd link test

Measure the actual radio path with the flight antenna, ground antenna and intended receiver configuration. Record packet success/loss, received signal behaviour, antenna orientation and system configuration. Treat a link-budget calculation as a model that must be checked by measurement.

3

INTEGRATED / DATA

to-end mission rehearsal

Run the complete payload and ground station together: acquisition -> timestamp -> local logging -> telemetry -> dashboard -> simulated loss of link -> recovery of stored data. Confirm that a communications interruption does not destroy the primary dataset.

VSWR = Voltage Standing Wave Ratio. It is an antenna/feed-system metric; a low value is desirable, but it does not by itself prove a long-range link.

For RF results, preserve frequency, bandwidth, antenna type, cable losses, power setting, distance, terrain and test conditions with the measurement.

19 -

Recovery is part of the design

Recovery is also a data-integrity problem: the mission is not complete if the payload lands safely but the recorded science cannot be retrieved or trusted.

1

MASS & CONFIGURATION

Weigh the complete flight train before the final configuration is frozen. Re-check mass after late hardware changes because mass affects lift, ascent rate, burst prediction and descent behaviour.

2

DESCENT & PARACHUTE

Ground-test the recovery system and confirm that the selected parachute/flight-train configuration produces an acceptable descent rate for the payload. Record test conditions and configuration.

3

FLIGHT TERMINATION / CUTDOWN

Verify the planned termination/cutdown function at a safe test level and confirm that its control logic, electrical state and manual fallback are documented. Do not treat a single trigger path as the only recovery safeguard.

4

LOCATOR CHAIN

Verify GPS position reporting, primary telemetry, secondary tracking where used, local logging, audible/visual locator behaviour and ground-station procedures. Simulate loss of one tracking path.

5

FIELD RECOVERY

Prepare the predicted landing area, maps, communications, personnel roles, transport and payload-identification procedure before launch. After recovery, preserve the original storage media and document any physical or environmental damage before powering or modifying the payload.

RECOVERY OUTPUT

Recovered payload + untouched raw data + flight log + configuration record + calibration metadata + anomaly/recovery notes = a dataset that can be audited and scientifically interpreted.

20 —

Mature teams prevent predictable failures

The workflow closes only when the payload, data, and recovery plan are all ready.

Area	Beginner failure	Mature-team practice
Sensor accuracy	Trust one uncalibrated reading	Calibrate, quantify uncertainty, cross-check
Cold environment	Battery sag and resets	Thermal tests, insulation, power margin
Telemetry	Bench success, flight packet loss	Antenna planning, link tests, local logging
Data	Corrupt SD card or wrong units	Redundant logging, watchdogs, timestamps
Integration	Subsystems fail when connected	Full system integration and rehearsal
Recovery	Launch plan without a find plan	Tracking roles, maps, chase vehicle

APPENDIX B / FLIGHT PHYSICS & RF LINK BUDGET

Appendix B — Flight Physics & RF Link Budget (35 km Case Study)

This appendix reproduces, in full, the subsystem-readiness calculation that justifies reaching 35 km with the hardware already chosen for IN-SPARX, and the two subsystem risks that need attention (thermal/power, and the BME280's altitude ceiling).

B.1 Slant Range at Burst Altitude

At burst altitude, the straight-line distance to the ground station is taken as essentially the altitude itself (35 km), since most of that distance is vertical rather than sideways for a near-overhead pass.

B.2 Free Space Path Loss (FSPL) at 868 MHz

$$FSPL\ (dB) = 20 \cdot \log_{10}(dkm) + 20 \cdot \log_{10}(fMHz) + 32.44$$

At 868 MHz and 35 km: $FSPL = 20 \cdot \log_{10}(35) + 20 \cdot \log_{10}(868) + 32.44 = 122.1\ dB$.

B.3 Link Budget with IN-SPARX Hardware

TERM	VALUE	MEANING
Pt	+20 dBm	SX1276 transmit power
Gtx	+2 dBi	Payload monopole/whip antenna gain
Grx	+9 dBi	Ground Yagi antenna gain
L cable	2 dB	Connectors and feedline loss
FSPL	122.1 dB	Free-space path loss at 35 km / 868 MHz
Pr (received power)	93.1 dBm	$20 + 2 + 9 - 2 - 122.1$
Smin (receiver sensitivity)	132 dBm	SX1276 datasheet class
Link margin	+38.9 dB	$Pr - Smin$

Conclusion: with the exact hardware already chosen (SX1276, 868 MHz, gi receiver), the link has almost 39 dB of spare margin at 35 km slant range.

APPENDIX B / THERMAL & INSTRUMENTATION

Thermal survival and the 10 km instrumentation ceiling

B.4 Thermal & Power Survival, 0–35 km

Between roughly 11 and 17 km the flight passes through the tropopause, where temperatures typically bottom out around 50 °C to 80 °C – the single harshest thermal moment of the whole 0–35 km profile.

The bigger practical risk, however, is the 3.7 V LiPo cell: a LiPo's internal resistance rises sharply as it gets cold, so under load the terminal voltage can sag enough to trip the 5 V boost converter's under-voltage cutoff – even though the battery still has plenty of stored charge.

Mitigation already built into the design: the EPS/XPS thermal hull. Recommendation: before the 35 km attempt, run the cold-chamber/freezer test described in Section 6.2 with the LiPo, boost converter and full electronics stack running together inside the actual EPS box.

THERMAL EXPOSURE / OPERATING RISK

11–17 km · tropopause crossing

50 °C to 80 °C · temperature minimum

3.7 V LiPo → voltage sag under load → 5 V boost-converter cutoff

B.5 Instrumentation Ceiling: Why 10 km Is the Right First Milestone

The Bosch BME280 has a datasheet-rated pressure range of approximately 300–1100 hPa, which corresponds to an altitude of roughly 9 km. Above roughly 9 km the BME280 will most likely stop producing meaningful pressure readings.

- Accept the gap: keep the BME280 for low-altitude data (0–9 km) and rely on GPS altitude and/or the ascent-rate model.
- Add a GPS module for altitude above 9 km.
- Add a wider-range pressure sensor.

Recommendation: log raw BME280 output for the whole flight regardless, but plan the official altitude-vs-time science profile above ~9 km around GPS and/or tl model, unless a wider-range sensor is added.

APPENDIX B / PHASE COMPARISON

What changes from 10 km to 35 km

B.6 What Changes Going From 10 km to 35 km — Summary

Overall conclusion: the physics and the hardware already chosen support reaching 35 km. The RF link and the thermal/power design do not need to change between the 10 km test and the final flight. The one subsystem that does need a decision before the final flight is instrumentation: how altitude and pressure will be measured above the BME280's ~9 km ceiling.

SUBSYSTEM	10 KM (PHASE 1)	35 KM (FINAL)
Balloon	Smaller / higher relative fill (deliberately burst low)	Larger (800-1200 g class), Free Lift tuned for ~5 m/s
RF link (SX1276)	~39+ dB margin	~39 dB margin
Avionics power/thermal (LiPo + EPS)	Same design; tropopause crossing already the critical test	Same design; same tropopause crossing, longer total time above it
Instrumentation (BME280)	Valid for nearly the whole flight	Loses validity above ~9 km; needs GPS or a wider-range sensor

Appendix C — Team Structure, Roles & Regulatory Notes

C.1 The IN-SPARX 7-Team Structure

Team 01 — Avionics Architecture	Team 05 — Software & Telemetry
Team 02 — Telemetry & RF Links	Team 06 — Atmospheric Science
Team 03 — Instrumentation Suite	Team 07 — Mission Control Ops
Team 04 — Structure & Thermal	

Each specialized team owns a technical domain, maintains documents, and delivers verified hardware/data handoffs.

APPENDIX C / ROLES & READINESS

Team structure at operating depth

C.2 Alternate Expanded Structure (Research & Planning Report)

Recommended core: 8–12 active students for a four-topic mission; smaller teams can work if members cover multiple roles. This is a practical inference for a student club, not an industry-standard average.

SUB-TEAM	SUGGESTED CORE PEOPLE	FIRST-MISSION DELIVERABLE
Atmospheric Physics + Weather	2	Temperature/pressure/humidity profile + atmospheric-model comparison
Optics	1–2	Light/scattering/imaging experiment with controlled exposure and calibration
Radiation	1–2	Detector characterization + altitude/time count dataset
Avionics & Power	2	Flight computer, sensors, battery/power budget, wiring and health checks
Telemetry & Ground Station	1–2	GPS/radio link, ground receiver, live tracking and local backup
Mechanical & Recovery	1–2	Payload enclosure, thermal protection, parachute and recovery plan
Software/Data	1–2	Firmware, logging, decoding, plots and scientific analysis
Mission Lead/Documentation	1	Schedule, checklists, reviews, permissions and final report

C.3 What Professional Balloon Programmes Add

- Formal requirements defined before buying components.
- Design reviews: Preliminary Design Review (PDR), Critical Design Review (CDR), Flight-Readiness Review (FRR).
- Calibration and uncertainty treated as part of the scientific result, not an afterthought.
- Environmental qualification: temperature, vibration/handling, power behaviour and data integrity tested before flight.
- Redundancy: important data and recovery functions never depend on a single tracker, sensor or storage device.
- Data integrity: raw data preserved, time-stamped, validated, and accompanied by sensor-configuration and calibration metadata.

A flight is only successful if the data can answer the original research question.

APPENDIX C / CLEARANCE & BUILD

Regulatory notes before launch

C.4 India-Specific Regulatory & Safety Notes

Free-flying balloons occupy controlled airspace and require coordination and clearance before launch. Balloon operations are governed by the Ministry of Civil Aviation, and flight information is communicated to the Airports Authority of India (AAI) in advance for clearance/NOTAM processing – the Master Plan’s own checklist recommends filing at least 14 days prior to launch.

The exact requirements for any specific IN-SPARX flight must be confirmed with the college, the relevant aviation authorities, and the launch-site authority before any launch is planned. This report, and the budgets and checklists within it, do not constitute launch authorization.

PRE-LAUNCH / VERIFY BEFORE PLANNING

Coordinate and confirm flight-specific requirements with the college, relevant aviation authorities, and launch-site authority.

Appendix D — Problems, Solutions & Technology Roadmap

This appendix walks the entire mission timeline – build, integration, launch day, and each altitude tier in turn – and lists the specific problem expected at that stage, why it happens, and the exact solution or technology from the IN-SPARX hardware/software stack that addresses it.

D.1 Build & Bench-Integration Problems

PROBLEM	WHY IT HAPPENS	SOLUTION / TECHNOLOGY
Loose wiring disconnects under vibration	Breadboard/jumper-wire connections are not vibration-rated	Mill a custom PCB with JST-XH locking connectors
Shared SPI bus conflicts	Both the SX1276 and the SD card use SPI	Follow the pin-by-pin interconnect blueprint exactly
Sensor reads corrupted by electrical noise	Long wire runs and shared rails pick up switching noise	Isolated I2C & SPI buses with decoupling capacitors
Firmware hangs or soft-locks mid-flight	Blocking code stalls the whole loop	Hardware Watchdog Timer and FreeRTOS multithreaded task scheduling
SD card writes stall or drop packets under load	Card write latency briefly blocks the main loop	Ring-buffer logging decouples SD writes from radio transmission
Mass creeps up late in the build	Extra wiring and additions accumulate	Set a mass budget from day one and weigh the train

APPENDIX D / FLIGHT OPERATIONS

Launch-day and altitude-tier problems

D.2 Launch-Day Problems

PROBLEM	WHY IT HAPPENS	SOLUTION / TECHNOLOGY
Weather scrubs or delays the launch window	Surface wind, cloud ceiling or upper-air wind shear makes the flight unsafe	Strict go/no-go criteria and CUSF Tawhiri predictor
Airspace/regulatory delay	NOTAM and civil-aviation clearance takes time to process	File NOTAM / airspace clearance at least 14 days prior
Imprecise helium fill – too little or too much free lift	Filling “by eye” changes climb speed and burst altitude	Precision helium-fill protocol using a calibrated weight
Payload disconnects or is damaged during handling at launch	Rigging not tension-tested or connectors not locked	50 kg shock-load rigging, JST-XH connectors and a final tug-test
Radio link drops immediately after release	Ground antenna not yet pointed at the payload or payload swing takes it through an antenna null	Rehearsed Yagi tracking and local logging
Chase team starts too late / heads the wrong direction	No pre-committed landing-zone estimate	Brief the Tawhiri-predicted landing zone before launch

D.3 Reaching 10 km — Problems & Solutions

The 10 km validation flight is deliberately the easiest tier, but it is the first time every subsystem operates together in flight conditions rather than on the bench.

- Balloon bursts earlier or later than the planned ~10 km: use the 4-step burst-altitude method with a 350–500 g balloon filled for a deliberately higher relative free lift.
- Electronics reset or behave erratically passing through the tropopause: use a 15 mm XPS thermal enclosure and cold-chamber test the full stack.
- GPS or LoRa telemetry gaps during ascent: use an independent LoRa + GPS + buzzer tracking stack and MicroSD backup.
- Parachute deployment issues at low altitude: ground-test deployment and inspect rigging before every flight.

D.4 Reaching 20 km — Problems & Solutions

At 20 km the flight spends longer in stratospheric cold and needs a second tracking link, since the balloon may drift further from the ground station.

- Longer cold-soak drains battery performance: add hand warmers alongside the 25 mm XPS enclosure.
- Payload drifts beyond APRS/LoRa range: add APRS (144.390 MHz) as a second independent tracking link alongside LoRa.
- Recovery team cannot spot the payload visually after landing: add a visual strobe/LED beacon that triggers below ~1000 m descent.
- BME280 pressure readings start to lose accuracy: rely on GPS altitude for the official profile above ~9 km.